Proof of the Benamou–Brenier Formula

Statement

Let

$$\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R}^d).$$

Then

$$W_2^2(\mu_0, \mu_1) = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathbb{R}^d} |v_t(x)|^2 \, d\rho_t(x) \, dt,$$

where the infimum is taken over all narrowly continuous curves

$$t \mapsto \rho_t \in \mathcal{P}_2(\mathbb{R}^d)$$

and Borel velocity fields $v_t \in L^2(\rho_t)$ satisfying the continuity equation

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0$$

in the sense of distributions, with endpoint conditions

$$\rho_0 = \mu_0, \qquad \rho_1 = \mu_1.$$

Proof

Denote the dynamic action by

$$\mathcal{A}(\rho, v) = \int_0^1 \int_{\mathbb{R}^d} |v_t(x)|^2 \, d\rho_t(x) \, dt.$$

We prove the equality by proving the two inequalities.

Step 1: Every admissible flow costs at least W_2^2

Let (ρ_t, v_t) be admissible and assume that $\mathcal{A}(\rho, v) < \infty$. The continuity equation says that the evolution of ρ_t can be realized by moving particles with Eulerian velocity v_t.

The analytic result behind this statement is the superposition principle: there exists a probability measure η on absolutely continuous paths

$$\omega : [0, 1] \to \mathbb{R}^d$$

such that

$$(e_t)_{\#}\eta = \rho_t, \qquad e_t(\omega) = \omega(t),$$

and, for η-almost every path,

$$\dot{\omega}(t) = v_t(\omega(t))$$

for almost every $t \in [0, 1]$.

Define the endpoint coupling

$$\gamma = (e_0, e_1)_{\#}\eta.$$

Since $(e_0)_{\#}\eta = \rho_0 = \mu_0$ and $(e_1)_{\#}\eta = \rho_1 = \mu_1$, we have

$$\gamma \in \Pi(\mu_0, \mu_1).$$

Therefore

$$W_2^2(\mu_0, \mu_1) \le \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 \, d\gamma(x, y).$$

Using the definition of γ, this becomes

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 \, d\gamma(x, y) = \int |\omega(1) - \omega(0)|^2 \, d\eta(\omega).$$

For each absolutely continuous path, Cauchy's inequality gives

$$|\omega(1) - \omega(0)|^2 = \left| \int_0^1 \dot\omega(t) \, dt \right|^2 \le \int_0^1 |\dot\omega(t)|^2 \, dt.$$

Integrating with respect to η yields

$$W_2^2(\mu_0, \mu_1) \le \int \int_0^1 |\dot\omega(t)|^2 \, dt \, d\eta(\omega).$$

Since $\dot\omega(t) = v_t(\omega(t))$ and $(e_t)_{\#}\eta = \rho_t$,

$$\int \int_0^1 |\dot\omega(t)|^2 \, dt \, d\eta(\omega) = \int_0^1 \int_{\mathbb{R}^d} |v_t(x)|^2 \, d\rho_t(x) \, dt.$$

Thus every admissible pair satisfies

$$W_2^2(\mu_0, \mu_1) \le \mathcal{A}(\rho, v).$$

Taking the infimum over all admissible pairs gives

$$W_2^2(\mu_0, \mu_1) \le \inf_{\rho, v} \mathcal{A}(\rho, v).$$

Step 2: An optimal transport plan realizes this cost

Let $\gamma \in \Pi(\mu_0, \mu_1)$ be an optimal coupling for the quadratic transport problem, so that

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 \, d\gamma(x, y) = W_2^2(\mu_0, \mu_1).$$

For each pair (x, y), move the particle from x to y along the straight line

$$X_t(x, y) = (1 - t)x + ty.$$

Define

$$\rho_t = (X_t)_{\#}\gamma.$$

This gives $\rho_0 = \mu_0$ and $\rho_1 = \mu_1$.

The Lagrangian velocity of the particle labelled by (x, y) is constant:

$$\frac{d}{dt} X_t(x, y) = y - x.$$

To express this motion in Eulerian form, disintegrate the labels (x, y) conditioned on the current position $z = X_t(x, y)$ and define

$$v_t(z) = \mathbb{E}_\gamma[y - x \mid X_t(x, y) = z].$$

Equivalently, v_t is the conditional average velocity of all particles that occupy the point z at time t.

For every smooth compactly supported test function φ, we compute

$$\frac{d}{dt} \int_{\mathbb{R}^d} \varphi(z)\, d\rho_t(z) = \frac{d}{dt} \int_{\mathbb{R}^d \times \mathbb{R}^d} \varphi(X_t(x, y))\, d\gamma(x, y).$$

By the chain rule,

$$\frac{d}{dt} \int \varphi(X_t(x, y))\, d\gamma(x, y) = \int \nabla\varphi(X_t(x, y)) \cdot (y - x)\, d\gamma(x, y).$$

By the definition of the conditional expectation v_t,

$$\int \nabla\varphi(X_t(x, y)) \cdot (y - x)\, d\gamma(x, y) = \int_{\mathbb{R}^d} \nabla\varphi(z) \cdot v_t(z)\, d\rho_t(z).$$

This is precisely the weak form of

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0.$$

Hence (ρ_t, v_t) is admissible.

It remains to estimate its action. By Jensen's inequality for conditional expectations,

$$|v_t(z)|^2 = |\mathbb{E}_\gamma[y - x \mid X_t = z]|^2 \le \mathbb{E}_\gamma[|y - x|^2 \mid X_t = z].$$

Integrating with respect to ρ_t gives

$$\int_{\mathbb{R}^d} |v_t(z)|^2\, d\rho_t(z) \le \int_{\mathbb{R}^d \times \mathbb{R}^d} |y - x|^2\, d\gamma(x, y).$$

Therefore

$$\mathcal{A}(\rho, v) \le \int_0^1 \int_{\mathbb{R}^d \times \mathbb{R}^d} |y - x|^2\, d\gamma(x, y)\, dt = W_2^2(\mu_0, \mu_1).$$

Taking the infimum over admissible pairs gives

$$\inf_{\rho, v} \mathcal{A}(\rho, v) \le W_2^2(\mu_0, \mu_1).$$

Conclusion

The first step gave

$$W_2^2(\mu_0, \mu_1) \le \inf_{\rho, v} \mathcal{A}(\rho, v),$$

while the second step gave

$$\inf_{\rho, v} \mathcal{A}(\rho, v) \le W_2^2(\mu_0, \mu_1).$$

Thus

$$W_2^2(\mu_0, \mu_1) = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathbb{R}^d} |v_t(x)|^2\, d\rho_t(x)\, dt.$$

Geometric Meaning

The static formulation chooses an optimal coupling between initial and final positions. The Benamou–Brenier formulation inserts time between those positions. It says that the squared Wasserstein distance is the least kinetic energy needed to deform μ_0 into μ_1 through a mass-preserving flow.